BRIGHT TUITION CLASSES

JEE Main

STD : 11 th

Date: 10/01/2026 **Maximum Marks: 300**

Timing: 5 pm to 8 pm

General Instructions

1. The test is of **3 hours** duration and the maximum marks is **300.**
2. The question paper consists of **3 Parts** (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).
3. **Section 1** contains **20 Multiple Choice Questions**. Each question has 4 choices **(A)**, **(B)**, **(C)** and **(D)**, out of which **ONLY ONE CHOICE** is correct.
4. **Section 2** contains **5 Numerical Value Type Questions**. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, ⊖ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)
5. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
7. On completion of the test, the candidate must hand over the Answer Sheet to the **Invigilator** on duty in the Room/Hall. **However, the candidates are allowed to take away this Test Booklet with them.**
8. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Marking Scheme

1. **Section – 1:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.
2. **Section – 2:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.

Name of the Candidate (In CAPITALS) : _____

Roll Number : _____

OMR Bar Code Number : _____

Candidate's Signature : _____ **Invigilator's Signature** _____

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Bright Tuition Class

Subject : Physics
Standard : 11
Total Mark : 100

Jee physics grand test std 11

Paper Set : 1
Date : 10-01-2026
Time : 3H:0M

Physics - Section A (MCQ)

(1) Two soap bubbles of radius $2\ cm$ and $4\ cm$, respectively, are in contact with each other. The radius of curvature of the common surface, in cm, is _____ .

(A) 4 (B) 3
(C) 2 (D) 1

(2) a $400\ g$ solid cube having an edge of length $10\ cm$ floats in water. How much volume of the cube is outside the water?
(Given : density of water $= 1000\ kgm^{-3}$)

(A) $1400\ cm^3$ (B) $4000\ cm^3$
(C) $400\ cm^3$ (D) $600\ cm^3$

(3) Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R
Assertion A : In a central force field, the work done is independent of the path chosen
Reason R : Every force encountered in mechanics does not have an associated potential energy. In the light of the above statements, choose the most appropriate answer from the options given below: –

(A) A is true but R is false
(B) Both A and R are true but R is NOT the correct explanation of A
(C) Both A and R are true and R is the correct explanation of A
(D) A is false but R is true

(4) Two particles are located at equal distance from origin. The position vectors of those are represented by $\vec{A} = 2\hat{i} + 3n\hat{j} + 2\hat{k}$ and $\vec{B} = 2\hat{i} - 2\hat{j} + 4p\hat{k}$, respectively. If both the vectors are at right angle to each other, the value of n^{-1} is _____ .

(A) 2 (B) 3
(C) 4 (D) 5

(5) A rod of linear mass density λ and length L is bent to form a ring of radius R. Moment of inertia of ring about any of its diameter is

(A) $\frac{\lambda L^3}{16\ \pi^2}$ (B) $\frac{\lambda L^3}{12}$
(C) $\frac{\lambda L^3}{4\ \pi^2}$ (D) $\frac{\lambda L^3}{8\ \pi^2}$

(6) A block of mass $1\ kg$, moving along x with speed $v_i = 10\ m/s$ enters a rough region ranging from $x = 0.1\ m$ to $x = 1.9\ m$. The retarding force acting on the block in this range is $F_r = -kx\ N$, with $k = 10\ N/m$. Then the final speed of the block as it crosses rough region is ..... m/s

(A) 10 (B) 4
(C) 6 (D) 8

(7) The position of a particle moving on $x-$axis is given by $x(t) = A\sin t + B\cos^2 t + Ct^2 + D$, where t is time. The dimension of $\frac{ABC}{D}$ is–

(A) L (B) $L^3\ T^{-2}$
(C) $L^2\ T^{-2}$ (D) L^2

(8) A tube of length $1\ m$ is filled completely with an ideal liquid of mass $2M$, and closed at both ends. The tube is rotated uniformly in horizontal plane about one of its ends. If the force exerted by the liquid at the other end is F then angular velocity of the tube is $\sqrt{\frac{F}{\alpha M}}$ in SI unit. The value of α is _____.

(A) 6 (B) 9
(C) 8 (D) 1

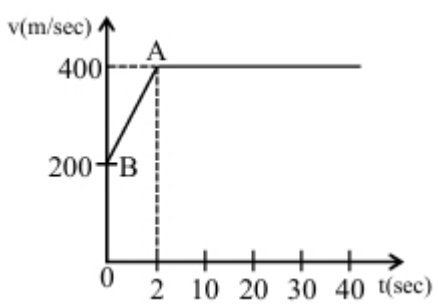
(9) A cubic block of mass m is sliding down on an inclined plane at $60°$ with an acceleration of $\frac{g}{2}$, the value of coefficient of kinetic friction is

(A) $\sqrt{3} - 1$ (B) $\frac{\sqrt{3}}{2}$
(C) $\frac{\sqrt{2}}{3}$ (D) $1 - \frac{\sqrt{3}}{2}$

(10) Two wires A and B are made of same material having ratio of lengths $\frac{L_A}{L_B} = \frac{1}{3}$ and their diameters ratio $\frac{d_A}{d_B} = 2$. If both the wires are stretched using same force, what would be the ratio of their respective elongations?

(A) $1 : 6$ (B) $1 : 12$
(C) $3 : 4$ (D) $1 : 3$

(11) The motion of an airplane is represented by velocity$-$time graph as shown below. The distance covered by airplane in the first 30.5 second is_____ km.



(A) 9 (B) 6
(C) 3 (D) 12

(12) A body of mass 4 kg experiences two forces $\vec{F}_1 = 5\hat{i} + 8\hat{j} + 7\hat{k}$ and $\vec{F}_2 = 3\hat{i} - 4\hat{j} - 3\hat{k}$. The acceleration acting on the body is :

(A) $-2\hat{i} - \hat{j} - \hat{k}$ (B) $4\hat{i} + 2\hat{j} + 2\hat{k}$
(C) $2\hat{i} + \hat{j} + \hat{k}$ (D) $2\hat{i} + 3\hat{j} + 3\hat{k}$

(13) A thin solid disk of $1\ kg$ is rotating along its diameter axis at the speed of $1800\ rpm$. By applying an external torque of $25\ \pi\ Nm$ for $40\ s$, the speed increases to $2100\ rpm$. The diameter of the disk is _____ m.

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(A) 10 (B) 20

(C) 30 (D) 40

(14) A satellite is launched into a circular orbit of radius $'R'$ around the earth. A second statellite is launched into an orbit of radius $1.03\ R$. The time period of revolution of the second satellite is larger than the first one approximately by : –

(A) 3% (B) 4.5%

(C) 9% (D) 2.5%

(15) A satellite of mass $\frac{M}{2}$ is revolving around earth in a circular orbit at a height of $\frac{R}{3}$ from earth surface. The angular momentum of the satellite is $M\sqrt{\frac{GMR}{x}}$. The value of x is _____ , where M and R are the mass and radius of earth, respectively. (G is the gravitational constant)

(A) 1 (B) 3

(C) 4 (D) 5

(16) A satellite of mass $1000\ kg$ is launched to revolve around the earth in an orbit at a height of $270\ km$ from the earth's surface. Kinetic energy of the satellite in this orbit is _____ $\times 10^{10}\ J$.
(Mass of earth $= 6 \times 10^{24}\ kg$, Radius of earth $= 6.4 \times 10^{6}\ m$, Gravitational constant $= 6.67 \times 10^{-11}\ Nm^2\ kg^{-2}$)

(A) 3 (B) 2

(C) 9 (D) 5

(17) An object of mass $'m'$ is projected from origin in a vertical xy plane at an angle $45°$ with the $x-$axis with an initial velocity v_0. The magnitude and direction of the angular momentum of the object with respect to origin, when it reaches at the maximum height, will be [g is acceleration due to gravity]

(A) $\frac{mv_0^3}{2\sqrt{2}g}$ along negative $z-$axis (B) $\frac{mv_0^3}{2\sqrt{2}g}$ along positive $z-$axis

(C) $\frac{mv_0^3}{4\sqrt{2}g}$ along positive $z-$axis (D) $\frac{mv_0^3}{4\sqrt{2}g}$ along negative $z-$axis

(18) Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A) : Time period of a simple pendulum is longer at the top of a mountain than that at the base of the mountain.
Reason (R) : Time period of a simple pendulum decreases with increasing value of acceleration due to gravity and vice$-$versa.
In the light of the above statements, choose the most appropriate answer from the options given below:

(A) Both (A) and (R) are true but (R) is not the correct explanation of (A). (B) Both (A) and (R) are true and (R) is the correct explanation of (A).

(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

(19) A uniform rod of mass $250\ g$ having length $100\ cm$ is balanced on a sharp edge at $40\ cm$ mark. A mass of $400\ g$ is suspended at $10\ cm$ mark. To maintain the balance of the rod, the mass to be suspended at $90\ cm$ mark, is

(A) $300\ g$ (B) $190\ g$

(C) $200\ g$ (D) $290\ g$

(20) Acceleration due to gravity on the surface of earth is 'g'. If the diameter of earth is reduced to one third of its original value and mass remains unchanged, then the acceleration due to gravity on the surface of the earth is _____ g.

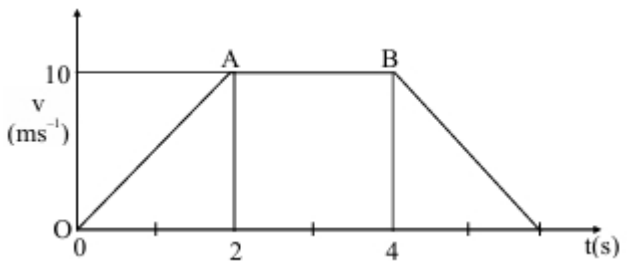
(A) 9 (B) 6

(C) 4 (D) 8

Physics - Section B (NUMERIC)

(21) The least count of a screw guage is $0.01\ mm$. If the pitch is increased by 75% and number of divisions on the circular scale is reduced by 50%, the new least count will be _____ $\times 10^{-3}\ mm$.

(22) The velocity$-$time graph of an object moving along a straight line is shown in figure. What is the distance (in m) covered by the object between $t = 0$ to $t = 4\ s$?



(23) A cricket player catches a ball of mass 120 g moving with 25 m/s speed. If the catching process is completed in 0.1 s then the magnitude of force exerted by the ball on the hand of player will be (in SI unit):

(24) A bullet is fired into a fixed target looses one third of its velocity after travelling 4 cm. It penetrates further D $\times 10^{-3}$ m before coming to rest. The value of D is:

(25) The moment of inertia of a solid disc rotating along its diameter is 2.5 times higher than the moment of inertia of a ring rotating in similar way. The moment of inertia of a solid sphere which has same radius as the disc and rotating in similar way, is n times higher than the moment of inertia of the given ring. Here, $n =$_____ Consider all the bodies have equal masses.

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Subject : Chemistry
Standard : 11
Total Mark : 100

11th Chemistry JEE (1 to 8)

Paper Set : 1
Date : 10-01-2026
Time : 0H:0M

Chemistry - Section A (MCQ)

(26) For electron in $'2s'$ and $'2p'$ orbitals, the orbital angular momentum values, respectively are :

(A) $\sqrt{2}\frac{h}{2\pi}$ and 0
(B) $\frac{h}{2\pi}$ and $\sqrt{2}\frac{h}{2\pi}$
(C) 0 and $\sqrt{6}\frac{h}{2\pi}$
(D) 0 and $\sqrt{2}\frac{h}{2\pi}$

(27) Match the LIST-I with LIST-II

	LIST-I		LIST-II
$A.$	Carbocation	$I.$	Species that can supply a pair of electrons.
$B.$	$C-$Free radical	$II.$	Species that can receive a pair of electrons.
$C.$	Nucleophile	$III.$	sp^2 hybridized carbon with empty p-orbital.
$D.$	Electrophile	$IV.$	sp^2/sp^3 hybridized carbon with one unpaired electron.

Choose the correct answer from the options given below :

(A) A-IV, B-II, C-III, D-I
(B) A-II, B-III, C-I, D-IV
(C) A-III, B-IV, C-II, D-I
(D) A-III, B-IV, C-I, D-II

(28) Among 10^{-9} g (each) of the following elements, which one will have the highest number of atoms?
Element : Pb, Po, Pr and Pt

(A) Po
(B) Pr
(C) Pb
(D) Pt

(29) Which of the following oxidation reactions are carried out by both $K_2Cr_2O_7$ and $KMnO_4$ in acidic medium?
$A.\ I^- \rightarrow I_2\ B.\ S^{2-} \rightarrow S\ C.\ Fe^{2+} \rightarrow Fe^{3+}\ D.\ I^- \rightarrow IO_3^-$
$E.\ S_2O_3{}^{2-} \rightarrow SO_4{}^{2-}$
Choose the correct answer from the options given below :

(A) B, C and D only
(B) A, D and E only
(C) A, B and C only
(D) C, D and E only

(30) The formation enthalpies, $\Delta H_f^{\ominus}$ for $H_{(g)}$ and $O_{(g)}$ are 220.0 and $250.0\ kJ\ mol^{-1}$, respectively, at $298.15K$, and ΔH_f^- for $H_2O_{(g)}$ is $-242.0\ kJ\ mol^{-1}$ at the same temperature. The average bond enthalpy of the $O-H$ bond in water at $298.15K$ is .....$kJ\ mol^{-1}$ (nearest integer).

(A) 466
(B) 566
(C) 666
(D) 766

(31) The equilibrium constant for decomposition of $H_2O(g)$
$H_2O(g) \rightleftharpoons H_2(g) + \frac{1}{2}O_2(g)\ \left(\Delta G^\circ = 92.34\ kJmol^{-1}\right)$ is 8.0×10^{-3} at $2300\ K$ and total pressure at equilibrium is 1 bar. Under this condition, the degree of dissociation (α) of water is ..... $\times 10^{-2}$(nearest integer value).
[Assume α is negligible with respect to 1]

(A) 1
(B) 2
(C) 3
(D) 5

(32) Given below are two statements :
Statement I : Hyperconjugation is not a permanent effect.
Statement II : In general, greater the number of alkyl groups attached to a positively charged $C-$atom, greater is the hyperconjugation interaction and stabilization of the cation.
In the light of the above statements, choose the correct answer from the options given below

(A) Statement I is true but Statement II is false
(B) Both Statement I and Statement II are false
(C) Statement I is false but Statement II is true
(D) Both Statement I and Statement II are true

(33) Among $SO_2, NF_3, NH_3, XeF_2, ClF_3$ and SF_4, the hybridization of the molecule with non$-$zero dipole moment and highest number of lone$-$pairs of electrons on the central atom is

(A) sp^3
(B) dsp^2
(C) $sp^3\ d^2$
(D) $sp^3\ d$

(34) The alkane from below having two secondary hydrogens is:

(A) $4-$Ethyl$-3,4-$dimethyloctane
(B) $2,2,4,4-$Tetramethylhexane
(C) $2,2,3,3-$Tetramethylpentane
(D) $2,2,4,5-$Tetramethylheptane

(35) Which one of the carbocations from the following is most stable?

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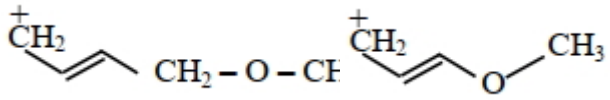
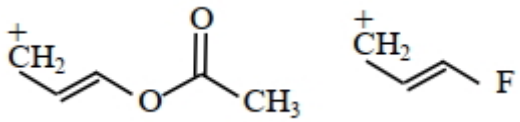
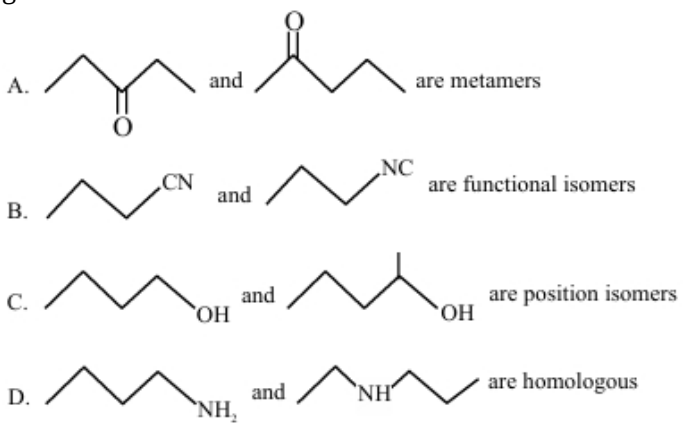
(A) (B)



(C) (D)



(36) Identify the correct statements from the following iamge Choose the correct answer from the options given below



(A) C & D only (B) B & C only

(C) A & B only (D) $A,\ B$ & C only

(37) The number of moles of methane required to produce 11 gCO_2(g) after complete combustion is:

(Given molar mass of methane in $gmol^{-1}$: 16)

(A) 0.75 (B) 0.25

(C) 0.35 (D) 0.5

(38) Match List-I with the List-II

List-I Reaction	List-II Type of redox reaction
(A) $N_{2(g)} + O_{2(g)} \rightarrow 2NO_{(g)}$	(I) Decomposition
(B) $2\ Pb\ (NO_3)_{2(s)} \rightarrow 2PbO_{(s)} + 4NO_{2(g)} + O_{2(g)}$	(II) Displacement
(C) $2Na_{(s)} + 2H_2O_{(l)} \rightarrow 2NaOH_{(aq.)} + H_{2(g)}$	(III) Disproportionation
(D) $2NO_{2(g)} + 2 - OH_{(aq)} \rightarrow NO^-_{2(aq.)} + NO^-_{3(qq.)} + H_2O_{(l)}$	(IV) Combination

Choose the correct answer from the options given below:

(A) $(A)-(I),(B)-(II),(C)-(III),(D)-(IV)$

(B) $(A)-(III),(B)-(II),(C)-(I),(D)-(IV)$

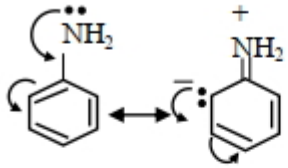
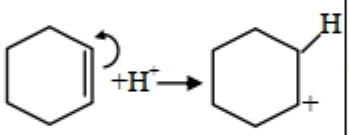
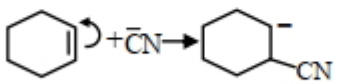
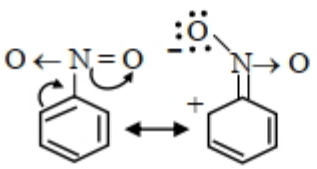
(C) $(A)-(II),(B)-(III),(C)-(IV),(D)-(I)$

(D) $(A)-(IV),(B)-(I),(C)-(II),(D)-(III)$

(39) Match List -I with List II:

(Image)

Choose the correct answer from the options given below :

List - I Mechanism steps		List - II Effect	
(A)		(I)	– E effect
(B)		(II)	– R effect
(C)		(III)	+ E effect
(D)		(IV)	+ R effect

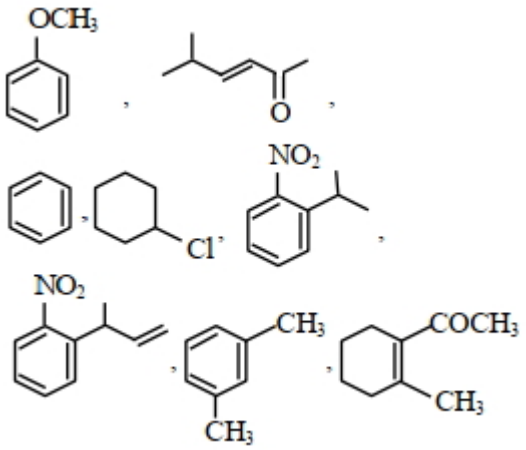
(A) $(A)-(IV),(B)-(III),(C)-(I),(D)-(II)$

(B) $(A)-(III),(B)-(I),(C)-(II),(D)-(IV)$

(C) $(A)-(II),(B)-(IV),(C)-(III),(D)-(I)$

(D) $(A)-(I),(B)-(II),(C)-(IV),(D)-(III)$

(40) How many compounds among the following compounds show inductive, mesomeric as well as hyperconjugation effects?



(A) 5 (B) 3

(C) 4 (D) 8

(41) Match List - I with List - II

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List - I (Pair of Compounds)	List - II (Isomerism)
An-propanol and Isopropanol	(I) Metamerism
BMethoxypropane and ethoxyethane	(II) Chain Isomerism
CPropanone and propanal	(III) Position Isomerism
DNeopentane and Isopentane	(IV) Functional Isomerism

Choose the correct answer from the options given below:

(A) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

(B) (A)-(III), (B)-(I), (C)-(II), (D)-(IV)

(C) (A)-(I), (B)-(III), (C)-(IV), (D)-(II)

(D) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

(42) The maximum number of orbitals which can be identified with n $= 4$ and $m_l = 0$ is

(A) 4 (B) 8

(C) 7 (D) 10

(43) Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A): Enthalpy of neutralisation of strong monobasic acid with strong monoacidic base is always -57 kJ mol^{-1}

Reason (R): Enthalpy of neutralisation is the amount of heat liberated when one mole of H^+ions furnished by acid combine with one mole of $^-$OH ions furnished by base to form one mole of water. In the light of the above statements, choose the correct answer from the options given below.

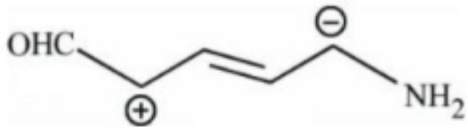
(A) (A) is true but (R) is false

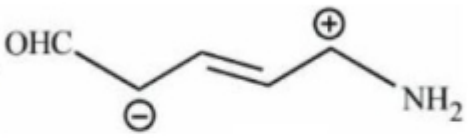
(B) Both (A) and (R) are true and (R) is the correct explanation of (A)

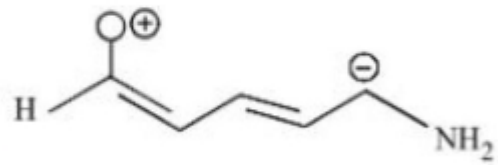
(C) (A) is false but (R) is true

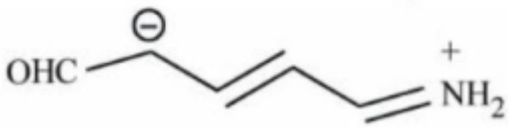
(D) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(44) Increasing order of stability of the resonance structure is:

A. 

B. 

C. 

D. 

(A) C, D, B, A (B) C, A, B, D

(C) D, C, A, B (D) D, C, B, A

(45) The number of atomic orbitals from the following having 5 radial nodes is ..... $7s, 7p, 6s, 8p, 8d$

(A) 3 (B) 2

(C) 1 (D) 4

Chemistry - Section B (NUMERIC)

(46) The energy of an electron in first Bohr orbit of $H-$atom is $-13.6\ eV$. The magnitude of energy value of electron in the first excited state of Be^{3+} is .....eV.(nearest integer value)

(47) Consider the following reaction occurring in the blast furnace.
$Fe_3O_{4(s)} + 4CO_{(g)} \rightarrow 3Fe_{(l)} + 4CO_{2(g)}$ ' x ' kg of iron is produced when $2.32 \times 10^3 kg\ Fe_3O_4$ and $2.8 \times 10^2 kgCO$ are brought together in the furnace. The value of ' x ' is .....(nearest integer)
Given : Molar mass of $Fe_3O_4 = 232g\ mol^{-1}$
Molar mass of $CO = 28g\ mol^{-1}$
Molar mass of $Fe = 56g\ mol^{-1}$

(48) Total number of non bonded electrons present in NO_2^-ion based on Lewis theory is.....

(49) In an atom, total number of electrons having quantum numbers n $= 4, |$ m$_l| = 1$ and m$_s = -\frac{1}{2}$ is

(50) 10 mL of gaseous hydrocarbon on combustion gives 40 mL of CO_2(g) and 50 mL of water vapour. Total number of carbon and hydrogen atoms in the hydrocarbon is.....

Bright Tuition Class

BRIGHT TUITIONS CLASSES

Subject : Mathematics
Standard : 11
Total Mark : 100

Ch 1 to 8

Paper Set : 1
Date : 10-01-2026
Time : 0H:0M

Mathematics - Section A (MCQ)

(51) In a certain town, 25% of the families own a phone and 15% own a car; 65% families own neither a phone nor a car and $2,000$ families own both a car and a phone. Consider the following three statements
(A) 5% families own both a car and a phone
(B) 35% families own either a car or a phone
(C) $40,000$ families live in the town
Then,
(A) Only (A) and (C) are correct.
(B) Only (B) and (C) are correct
(C) All (A), (B) and (C) are correct.
(D) Only (A) and (B) are correct

(52) If $A = \{x \in R : |x| < 2\}$ and $B = \{x \in R : |x-2| \geq 3\}$ then
(A) $A \cup B = R - (2,5)$ (B) $A \cap B = (-2,-1)$
(C) $B - A = R - (-2,5)$ (D) $A - B = [-1,2)$

(53) A survey shows that 63% of the people in a city read newspaper A whereas 76% read newspaper B. If $x\%$ of the people read both the newspapers, then a possible value of x can be
(A) 65 (B) 37
(C) 29 (D) 55

(54) The range of the function $f(x) = \sqrt{3-x} + \sqrt{2+x}$ is
(A) $[\sqrt{5}, \sqrt{10}]$ (B) $[2\sqrt{2}, \sqrt{11}]$
(C) $[\sqrt{5}, \sqrt{13}]$ (D) $[\sqrt{2}, \sqrt{7}]$

(55) Let $f(x) = \frac{1}{7-\sin 5x}$ be a function defined on R. Then the range of the function $f(x)$ is equal to:
(A) $\left[\frac{1}{8}, \frac{1}{5}\right]$ (B) $\left[\frac{1}{7}, \frac{1}{6}\right]$
(C) $\left[\frac{1}{7}, \frac{1}{5}\right]$ (D) $\left[\frac{1}{8}, \frac{1}{6}\right]$

(56) The sum of the solutions $x \in \mathbb{R}$ of the equation $\frac{3\cos 2x + \cos^3 2x}{\cos^6 x - \sin^6 x} = x^3 - x^2 + 6$ is
(A) 0 (B) 1
(C) -1 (D) 3

(57) The value of $(\sin 70^\circ)(\cot 10^\circ \cot 70^\circ - 1)$ is
(A) 1 (B) 0
(C) $3/2$ (D) $2/3$

(58) If $5\left(\tan^2 x - \cos^2 x\right) = 2\cos 2x + 9$, then $\cos 4x$ is equal to
(A) $-\frac{7}{9}$ (B) $-\frac{3}{5}$
(C) $\frac{1}{3}$ (D) $\frac{2}{9}$

(59) If $z = 2 + 3i$, then $z^5 + (\overline{z})^5$ is equal to.
(A) 244 (B) 224
(C) 245 (D) 265

(60) If for $z = \alpha + i\beta, |z+2| = z + 4(1+i)$, then $\alpha + \beta$ and $\alpha\beta$ are the roots of the equation
(A) $x^2 + 7x + 12 = 0$ (B) $x^2 + 3x - 4 = 0$
(C) $x^2 + 2x - 3 = 0$ (D) $x^2 + x - 12 = 0$

(61) Let $A = \left\{\theta \in (0, 2\pi) : \frac{1+2i\sin\theta}{1-i\sin\theta}\right.$ is purely imaginary $\}$. Then the sum of the elements in A is
(A) π (B) 2π
(C) 4π (D) 3π

(62) If all the words, with or without meaning, are written using the letters of the word $QUEEN$ and are arranged as in English dictionary, then the position of the word $QUEEN$ is
(A) 44th (B) 45th
(C) 46th (D) 47th

(63) A boy needs to select five courses from 12 available courses, out of which 5 courses are language courses. If he can choose at most two language courses, then the number of ways he can choose five courses is
(A) 454 (B) 465
(C) 546 (D) 645

(64) The number of triangles whose vertices are at the vertices of a regular octagon but none of whose sides is a side of the octagon is
(A) 24 (B) 56
(C) 16 (D) 48

(65) The remainder on dividing 5^{99} by 11 is
(A) 9 (B) 18
(C) 27 (D) 36

(66) The Coefficient of x^{-6}, in the expansion of $\left(\frac{4x}{5} + \frac{5}{2x^2}\right)^9$, is .....
(A) 5041 (B) 5042
(C) 5043 (D) 5040

(67) Let m and n be the coefficients of seventh and thirteenth terms respectively in the expansion of $\left(\frac{1}{3}\mathrm{x}^{\frac{1}{3}} + \frac{1}{2\mathrm{x}^{\frac{2}{3}}}\right)^{18}$. Then $\left(\frac{\mathrm{n}}{\mathrm{m}}\right)^{\frac{1}{3}}$ is :
(A) $\frac{4}{9}$ (B) $\frac{1}{9}$
(C) $\frac{1}{4}$ (D) $\frac{9}{4}$

(68) The sum of first 20 terms of the sequence $0.7, 0.77, 0.777, \ldots$ is
(A) $\frac{7}{18}\left(179 - 10^{-20}\right)$ (B) $\frac{7}{9}\left(99 - 10^{-20}\right)$
(C) $\frac{7}{81}\left(179 + 10^{-20}\right)$ (D) $\frac{7}{9}\left(99 + 10^{-20}\right)$

(69) The sum of all those terms, of the anithmetic progression $3, 8, 13, \ldots\ldots 373$, which are not divisible by 3,is equal to .....
(A) 9524 (B) 9523
(C) 9522 (D) 9525

(70) The roots of the quadratic equation $3x^2 - px + q = 0$ are 10th and 11th terms of an arithmetic progression with common difference $\frac{3}{2}$. If the sum of the first 11 terms of this arithmetic progression is 88 , then $q - 2p$ is equal to_____

(A) 474 (B) 426

(C) 423 (D) 478

.... Mathematics - Section B (NUMERIC) ....

(71) If the domain of the function $f(x) = \frac{\sqrt{x^2-25}}{(4-x^2)} + \log_{10}\left(x^2 + 2x - 15\right)$ is $(-\infty, \alpha)U[\beta, \infty)$, then $\alpha^2 + \beta^3$ is equal to :

(72) If $\sin x = -\frac{3}{5}$, where $\pi < x < \frac{3\pi}{2}$ then $80\left(\tan^2 x - \cos x\right)$ is equal to :

(73) If the center and radius of the circle $\left|\frac{z-2}{z-3}\right| = 2$ are respectively (α, β) and γ, then $3(\alpha + \beta + \gamma)$ is equal to

(74) The number of integers greater than 6000 that can be formed , using the digits $3, 5, 6, 7,$ and 8 without repetition is :

(75) The remainder when $7^{2022} + 3^{2022}$ is divided by 5 is.